ITBDA4-14 Project 2

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Course: Big Data Analytics Techniques Assessments
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Table of contents

1. Project
2. Instructions to Students
3. Question 1
4. Question 2
5. Question 3
6. Question 4

1. Project

Faculty:	Information Technology
Module Code:	ITBDA4-14
Module Name:	Big Data Analytics
Content Writer:	Dr Yves Matanga
Internal Moderation:	Community of Practice
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Total Marks:	100

Submission Week:	Week 6

This module is presented on NQF level 8

5% will be deducted from the student's assignment mark for each calendar day the assignment is submitted late, up to a maximum of three calendar days. The penalty will be based on the official campus submission date.

Projects submitted later than three calendar days after the deadline or not submitted will get 0%. [1]

This is an individual Project

Groups should consist of [Enter num.] members.

This Choose an item: project contributes 20% towards the final mark.

[1] Under no circumstances will assignments be accepted for marking after the assignments of other students have been marked and returned to the students.

2. Instructions to Students

1. Please ensure that your answer file (where applicable) is named as follows before submission: **Module Code – Assessment Type – Campus Name – Student Number.**
2. Remember to keep a copy of all submitted assignments.
3. All work must be typed.
4. Please note that you will be evaluated on your writing skills in all your assignments.
5. All work must be submitted through Turnitin. The full originality report will be automatically generated and available for the lecturer to assess. Negative marking will be applied if you are found guilty of plagiarism, poor writing skills, or if you have applied incorrect or insufficient referencing. (See the "instructions to students" book activity before this activity where the application of negative marking is explained.)
6. You are not allowed to offer your work for sale or to purchase the work of other students. This includes the use of professional assignment writers and websites, such as Essay Box. You are also not allowed to make use of artificial intelligence tools, such as ChatGPT, to create content and submit it as your own work. If this should happen, Eduvos reserves the right not to accept future submissions from you.

3. Question 1

Question 1 — 25 Marks

Study the scenario and complete the questions that follow:

MapReduce Process

The MapReduce framework is a pioneering methodology for handling large-scale data processing across multiple computers or data nodes, designed to tackle the challenges posed by the era of big data. As a new big data analyst, you have been tasked with developing a Java-based Hadoop MapReduce program. Your objective is to **search for a specific keyword** within a large corpus of documents and return the **row and column location of the keyword** in each document. For this task, you may use the word "**africa**" as the keyword for demonstration purposes. Ensure that your search is case-insensitive. A database of news article is provided as your corpus (**map_reduce_test_docs.zip**).

Source: Matanga, N.Y. (2026)

1.1. With the aid of a logical layout diagram, explain how you will perform the keyword search operation into the MapReduce framework and yield the expected result.

(7 marks)

Implement practically, a Map Reduce process with demonstration evidence and

1.2. Provide a R or Java implementation of the Mapper for the above problem.

(9 marks)

1.3. Provide a R or Java implementation of the Reducer class

(5 Marks)

1.4. Provide an implementation of the Hadoop MapReduce configuration settings

(4 marks)

Note: The keyword of search should be input dynamically and not hardcoded into the program. Submit a functional R or Java project for the MapReduce program.

End of Question 1

4. Question 2

Question 2 — 20 Marks

2.1. Given a collection of news files (**news.zip**), generate a data frame (col_1: file_name, col_2: content). Load the new content in each row and preprocess using appropriate data denoising techniques.

(5 marks)

2.2. Using Latent Dirichlet Allocation (LDA), find the top n topics in the documents and display the list of top k words per topic including appropriate reporting of word distribution per topic.

(9 marks)

2.3. Provide appropriate visualization for the top key words in each topic using at least two visualization methods.

(6 marks)

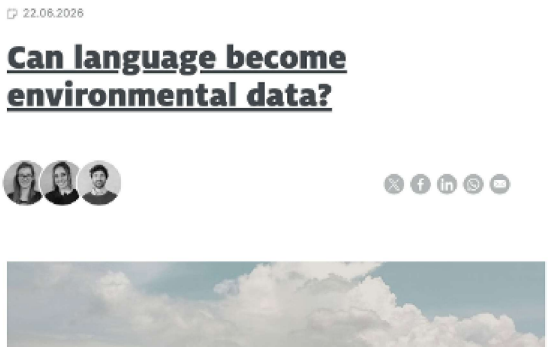
End of Question 2

5. Question 3

Question 3 — 30 Marks

Study the scenario and complete the questions that follow:

Building a WebText API – Advancing Research and Analysis



NLP is gaining momentum in Artificial Intelligence research with the advent of more and more sophisticated algorithms like large language models. An analytics company would like to advantage of this trend by building a Web-search automatic text mining API that would read data from a mere "web link" and generate insights from its content. As a maturing data scientist in this domain, you have been contacted to provide a proof of concept of the design.

Source: Matanga, N.Y. (2026)

3.1. Select a website of your own choosing with compliance of 'fair use of webscraping' and extract the clean text web content from the site leading to the formation of an R data structure.

(7 marks)

3.2. Perform textual data preprocessing using all appropriate techniques that clean the data content of non-pertinent information for
analysis.

(5 marks)

3.3. Provide text summarisation of the collected information using at least two text summarisation techniques including at least two
visualisations.

(18 marks)

All graphs must be fully labelled, and the findings must be well described. Include your R scripts to the narrative.

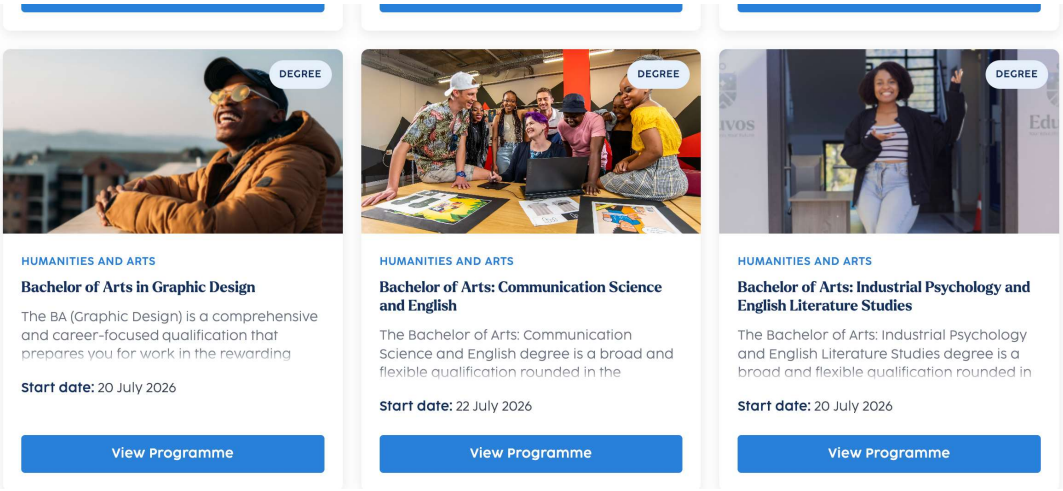
End of Question 3

6. Question 4

Question 4 — 25 Marks

Study the scenario and complete the questions that follow:

ETL manipulating using R & Postgres



Extract, Transform, and Load (ETL) is a fundamental methodology in data analytics for converting raw, unstructured data into structured datasets suitable for analysis. As a Data Analyst, you have been tasked with demonstrating the complete ETL workflow by extracting data from an unstructured, web-hosted dataset, transforming and cleaning the data, and loading it into a structured database to support analytical reporting.

Source: Matanga, N.Y. (2026)

4.1. Using web scraping and regular expression (regex) analysis, define a use case of your own choosing. Develop an R program that scrapes at least 20 unstructured records (e.g., articles, news posts, or blog entries) from a website and constructs a data frame containing at least both a textual and a numerical (existing or inferred) attribute. The extracted data should initially contain spurious, inconsistent, or incomplete information representative of real-world web data

(8 marks)

4.2. Apply appropriate data cleaning, denoising, and missing-value imputation techniques to improve the quality of the scraped dataset. Export the cleaned dataset as a CSV file suitable for further analysis

(4 marks)

4.3. Using PostgreSQL, import the cleaned CSV dataset into a relational table. Perform two relevant SQL-based analytical investigations on the data. Present the SQL queries together with a concise interpretation of the results obtained.

(7 marks)

4.4. Create a GitHub repository containing:

- The project documentation (Questions 1-4)
- The project datasets
- The Notebooks showcasing the flow of analytics for all questions 1 to 4

(6 marks)

End of Question 4